

A High-Low Ratio Test for Geometric Brownian Motion

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Abstract

We propose a test of whether security prices follow a geometric Brownian motion (gBm), based on HLR, the ratio of the average two-day to average one-day squared logarithm of high/low prices, and the associated test Z_{HLR} . We show that HLR has a much smaller asymptotic standard error than that of the corresponding variance ratio, VR, that tests for a random walk. We find in simulations that Z_{HLR} has much higher power than the corresponding VR test. In an empirical analysis of cryptocurrencies we find that in some cases Z_{HLR} yields statistically significant results whereas the variance ratio test does not.

Preliminary Version. Comments are Welcome.

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1 Introduction

Research in finance assumes that in a rational, efficient market, asset prices (in logarithms) follow a continuous-time stochastic process characterized by independent and identically distributed Gaussian-distributed increments. The process is described as a geometric Brownian motion, denoted gBm. Under the gBm, the (logarithmic) price is a random walk process and returns – the price increments – are serially uncorrelated.

Lo and Mackinlay (1988) propose a variance ratio test, denoted VR, of whether market prices (in logarithms), observed at discrete, equally-spaced times, follow a random walk. Employing returns (the difference of log prices), VR is calculated as the ratio of the sample variance of k -period returns to k times the variance of the one-period returns. The test is based on the extent to which VR differs from unity. Rejecting the null hypothesis suggests that the price process is not a gBm, the violation presumably stemming from returns not being serially uncorrelated.

This paper proposes a more general test of whether a price process follows a gBm, employing the highest and lowest prices observed for the process for certain intervals of continuous time. For a k -period interval, a value HL is the squared logarithm of the ratio of the highest and lowest prices during that interval. Our measure HLR, the high/low ratio, is the ratio of two values: The average of the values HL calculated over k -period intervals using the highest and lowest prices for that interval, and k times the average of the values HL calculated for single periods. For example, for $k = 2$, HLR is the ratio of the sample average squared logarithm of HL that employs the highest and lowest prices over the two-day intervals to twice the average squared logarithm of the daily ratio of the daily high and low prices.

Our simulations show that while HLR is close to unity under a process that resembles a gBm, it differs substantially from unity if the underlying price process deviates from the gBm. Deviations may arise from a nonzero return autocorrelation, or from the return distribution having fatter tails

than those of a Gaussian distribution. The latter type of deviation is not detected by VR but is detected by HLR.

For testing the null hypothesis of gBm, we develop a statistic based on asymptotic theory. Because Charles and Darne (2009, p. 504) point out that because " ... virtually nothing is often known about the exact distribution of the VR test statistic that uses overlapping data ... asymptotic distribution instead of exact distribution is often used for conducting statistical inference on the VR test." We show that the asymptotic variance of the distribution of our test statistic under the null is much smaller than that of a comparable VR test statistic proposed by Lo and MacKinlay (1988), making our HLR test potentially more powerful than the VR test. The greater power of our test reduces the necessity of using overlapping intervals which Lo and MacKinlay (1988) use in order to improve the power of their VR test.¹ Notably, the HLR test has power against certain alternatives that the VR does not test. This is partly because the tested hypotheses are not the same: the HLR tests for deviations from the gBm while the VR tests for whether prices behave as random walk. If the log price process is a gBm, log prices will behave as a random walk.

It should be recognized, however, that the variance calculated from the daily and bi-daily high and low prices is based on more information than that used for the calculation of the standard variance of the one-day and two-day returns. This has been recognized by Parkinson (1980) who noted that the return variance estimation by the extreme values method is superior to the standard method that employed squared returns that use only closing prices.²

We provide empirical analysis of the VR and HLR tests to study the price process of Bitcoin and Ethereum, the two major cryptocurrencies which trade continuously. We find that the HLR test detects significant deviations from the gBm in some sample periods. Using the same data for

¹Charles and Darne (2009) discuss problems of the VR test statistic that uses overlapping data.

²Parkinson (1980, p. 63) suggests that the random walk hypothesis could be tested using the log of the high/low price ratio, but he does not develop a test statistic, nor does his (implied) test involve the ratio of the average squared log of the high/low price ratios over different time periods.

tests based on the VR we find that there too there are sometimes rejections of the hypothesis that the price process is a gBm. As we suggest, while the VR and the HLR have a similar null hypothesis with regard to return autocorrelation, HLR enables to detect other deviations from gBm which the VR does not. Altogether, our HLR test provides a valuable tool in detecting violations from the gBm process. In what follows, we present in Section 2 the definition of our HLR measure. In Section 3 we define the version of the VR test that we will use in this paper. In Section 4 we develop the asymptotic test for gBm based on HLR. In Section 5 we present some theory on Brownian motion that is relevant for developing our asymptotic test. In Section 6 we present simulation results. In Section 7 we apply our HLR test and the VR test to study the daily price series of two leading cryptocurrencies, Bitcoin and Ethereum, which trade continuously without being subject to any regulatory constraints. We find cases with significant deviations from gBm.

2 Definition of HLR

Let $P(t)$ denote the continuous-time asset price at time $t \in [0, \infty)$, where t is in units of days. In discrete time $t = 1, 2, \dots$ we obtain the time series of daily asset returns,

$$r_t = \log P(t) - \log P(t-1) \quad , \quad t = 1, 2, 3, \dots \quad ,$$

where $\log$ denotes the natural log. Suppose for a particular asset we have observed $P(t)$ for $t \in [0, n]$ and therefore we have also observed the n daily stock returns $r_1, \dots, r_n$. For simplicity, we assume that n , the total number of daily returns, is even. We define HLR , the High/Low Ratio statistic, assuming non-overlapping two-day periods in the numerator (see below). Alternative versions could be considered based on three-day periods, four-day periods, etc., but we focus here on two-day periods. For even-numbered days, $t = 2, 4, \dots, n$, let $HH(t)$ and $LL(t)$ denote the high and low for the price P over the two-day period $(t-2, t]$, that is, the period comprising day $t-1$ and

day t . Thus,

$$HH(t) = \sup_{u \in (t-2, t]} P(u) \ , \ LL(t) = \inf_{u \in (t-2, t]} P(u) \ .$$

For any (even or odd-numbered) day, let $H(t)$ and $L(t)$ denote the daily high and low for day t ,

$$H(t) = \sup_{u \in (t-1, t]} P(u) \ , \ L(t) = \inf_{u \in (t-1, t]} P(u) \ .$$

Now, we define HLR , the high-low ratio, as

$$HLR = \frac{Ave \log^2[HH(t)/LL(t)]}{Ave\{\log^2[H(t)/L(t)] + \log^2[H(t-1)/L(t-1)]\}} \ , \quad (1)$$

where in both the numerator and denominator, Ave denotes the average of the values for $t = 2, 4, \dots, n$, and $\log^2$ denotes the square of the natural log.

3 Variance Ratio Statistic

We assume that there are n observed returns, $r_1, \dots, r_n$, where n is even. The variance ratio statistic is defined by

$$VR = \frac{v\hat{a}r(r_2 + r_1, r_4 + r_3, \dots, r_n + r_{n-1})}{2 v\hat{a}r(r_1, r_2, \dots, r_n)}$$

where for any sequence $x_1, \dots, x_m$,

$$v\hat{a}r(x_1, \dots, x_m) = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2 \ ,$$

and

$$\bar{x} = \frac{1}{m} \sum_{i=1}^m x_i \ .$$

4 Asymptotic Theory for HLR under gBm

In this section, we assume the null hypothesis that $P(t)$ is geometric Brownian motion (gBm) so that $\log P(t) = B(t)$, where $B(t)$ is ordinary Brownian motion. We could assume that $var B(1) = \sigma^2$

but σ^2 will cancel in the numerator and denominator of HLR ; without loss of generality we assume that $\text{var}B(1) = 1$, so $B(t)$ is standard Brownian motion. We derive the asymptotic distribution of $\sqrt{n}(HLR - 1)$ as $n \rightarrow \infty$. Since the asymptotic theory for HLR under the null hypothesis is based on the properties of Brownian motion, we consider a standard Brownian motion on $[0, 2]$, which represents the continuous-time log price on $[0, 2]$. Let h_1, ℓ_1, c_1 be, respectively the high low and close for the Brownian motion on $[0, 1]$ and let h_2, ℓ_2 be the high and low on $[1, 2]$. Let $\Delta_1 = h_1 - \ell_1$ and $\Delta_2 = h_2 - \ell_2$, the ranges on $[0, 1]$ and $[1, 2]$. Next, define

$$\Delta_{12} = \max(h_1, h_2) - \min(\ell_1, \ell_2) \ ,$$

as the range on $[0, 2]$. We have the following theorem.

Theorem 1 *Under the gBm null hypothesis,*

$$\sqrt{n}(HLR - 1) \xrightarrow{D} N\left(0, \frac{\sigma_{Num}^2}{32 \log^2 2}\right) \ , \quad (2)$$

where

$$\sigma_{Num}^2 = \text{var}[\Delta_{12}^2 - (\Delta_1^2 + \Delta_2^2)] \ .$$

In Section 5, we take steps towards obtaining an analytical expression for σ_{Num}^2 as a function of certain integrals. See, for example, Equation (5) which gives $E[\Delta_{12}^2]$ as a five-dimensional integral. So far, we have been unable to obtain numerically stable evaluations of this and other related integrals. In lieu of such evaluations, we have obtained a Monte Carlo approximation to σ_{Num}^2 by simulation of 1.5 million realizations of a standard Brownian motion on $[0, 2]$. This led to the approximate value $\tilde{\sigma}_{Num}^2 = 4.23$. The standard error for this Monte Carlo average is 0.009. In our subsequent data analysis and simulations we will perform the HLR test of the null hypothesis of gBm using the test statistic

$$Z_{HLR} = \frac{\sqrt{n}(HLR - 1)}{\sqrt{(4.23)/[(32) \log^2 2]}} = \frac{\sqrt{n}(HLR - 1)}{0.525} \ , \quad (3)$$

in a two-tailed test based on critical values from the standard normal distribution.

Proof of Theorem 1:

Note that

$$HLR - 1 = \frac{Ave \log^2[HH(t)/LL(t)] - Ave\{\log^2[H(t)/L(t)] + \log^2[H(t-1)/L(t-1)]\}}{Ave\{\log^2[H(t)/L(t)] + \log^2[H(t-1)/L(t-1)]\}} .$$

Furthermore,

$$H(t) = \sup_{u \in (t-1, t]} P(u), \quad L(t) = \inf_{u \in (t-1, t]} P(u), \quad \log H(t) = \sup_{u \in (t-1, t]} B(u), \quad \log L(t) = \inf_{u \in (t-1, t]} B(u).$$

Similarly,

$$HH(t) = \sup_{u \in (t-2, t]} P(u), \quad LL(t) = \inf_{u \in (t-2, t]} P(u), \quad \log HH(t) = \sup_{u \in (t-2, t]} B(u), \quad \log LL(t) = \inf_{u \in (t-2, t]} B(u).$$

Thus,

$$HLR - 1 \stackrel{D}{=} \frac{A_{2,n} - A_{1,n}}{A_{1,n}}$$

where

$$A_{2,n} = Ave\{[\sup_{u \in (t-2, t]} B(u) - \inf_{u \in (t-2, t]} B(u)]^2\}$$

and

$$A_{1,n} = Ave\{[\sup_{u \in (t-2, t-1]} B(u) - \inf_{u \in (t-2, t-1]} B(u)]^2\} + Ave\{[\sup_{u \in (t-1, t]} B(u) - \inf_{u \in (t-1, t]} B(u)]^2\} .$$

By ergodicity, as $n \rightarrow \infty$, $A_{1,n}$ converges almost surely to $E[\Delta_1^2] + E[\Delta_2^2]$. It was shown by Feller (1951) that for a standard Brownian motion on $[0, N]$ the range $R(N)$ satisfies

$$E[R(N)] = 2\sqrt{2N/\pi} \quad , \quad var[R(N)] = 4N(\log 2 - 2/\pi) .$$

Thus, $E[\Delta_1^2] = E[\Delta_2^2] = 4(\log 2 - 2/\pi) + 8/\pi = 4\log 2$ and $E[\Delta_{12}^2] = 8(\log 2 - 2/\pi) + 16/\pi = 8\log 2 = E[\Delta_1^2] + E[\Delta_2^2]$.

It suffices to derive the limiting distribution of $\sqrt{n/2}(A_{2,n} - A_{1,n})$. Note that $E[A_{2,n} - A_{1,n}] = E[\Delta_{12}^2] - (E[\Delta_1^2] + E[\Delta_2^2]) = 0$. We have $A_{2,n} = Ave\{X_{2,t}\}$ and $A_{1,n} = Ave\{X_{1,t}\}$ where

$$X_{2,t} = \left[\sup_{u \in (t-2, t]} B(u) - \inf_{u \in (t-2, t]} B(u) \right]^2$$

and

$$X_{1,t} = \left[\sup_{u \in (t-2, t-1]} B(u) - \inf_{u \in (t-2, t-1]} B(u) \right]^2 + \left[\sup_{u \in (t-1, t]} B(u) - \inf_{u \in (t-1, t]} B(u) \right]^2 .$$

Note that the $X_{2,t}$ are *iid*, the $X_{1,t}$ are *iid* and the two terms in $X_{1,t}$ are mutually independent, for $t = 2, 4, \dots, n$. It follows that the $X_{2,t} - X_{1,t}$ are *iid* with zero mean (and they have finite variance), so by the Central Limit Theorem,

$$\sqrt{n/2}(A_{2,n} - A_{1,n}) \xrightarrow{D} N(0, \sigma_{Num}^2) ,$$

where

$$\sigma_{Num}^2 = var(X_{2,t} - X_{1,t}) = var[\Delta_{12}^2 - (\Delta_1^2 + \Delta_2^2)] .$$

Overall, we have

$$\sqrt{n/2}(HLR - 1) \xrightarrow{D} N(0, \sigma_{Num}^2 / (8 \log 2)^2)$$

so

$$\sqrt{n}(HLR - 1) \xrightarrow{D} N\left(0, \frac{\sigma_{Num}^2}{32 \log^2 2}\right) . \square$$

5 Densities for relevant functions of Brownian motion on $[0, 2]$

Consider a standard Brownian motion on $[0, 2]$. Let h_1, ℓ_1, c_1 be the high, low and close on $[0, 1]$ and let h_2, ℓ_2 be the high and low on $[1, 2]$. Let $\Delta_1 = h_1 - \ell_1$ and $\Delta_2 = h_2 - \ell_2$, the ranges on $[0, 1]$ and $[1, 2]$. At time 1, if we subtract c_1 from the Brownian motion, the path on $[1, 2]$ is a Brownian motion, independent of the path on $[0, 1]$. From Corollary 2.3 of Riedel (2021), with $\sigma^2 = 1$, the

density of h_1, ℓ_1, c_1 is

$$p(h_1, \ell_1, c_1) = 4 \sum_{k=-\infty}^{\infty} \{k^2 a_k(c_1, \Delta_1) \phi(c_1 - 2k\Delta_1) - k(k+1) a_k(c_1 - 2h_1, \Delta_1) \phi(c_1 - 2h_1 - 2k\Delta_1)\}$$

where $a_k(c_1, \Delta_1) = (c_1 - 2k\Delta_1)^2 - 1$. From Corollary 2.4 of Riedel (2021), with $\sigma^2 = 1$, the density of h, ℓ , the high and low on $[0, 1]$, is

$$\begin{aligned} p(h, \ell) = & -4 \sum_{k=-\infty}^{\infty} k^2 [(h - 2k\Delta) \phi(h - 2k\Delta) - (\ell - 2k\Delta) \phi(\ell - 2k\Delta)] \\ & + 4 \sum_{k=-\infty}^{\infty} \{k(k+1) [(h - 2k\Delta - 2h) \phi(h - 2k\Delta - 2h) - (\ell - 2k\Delta - 2h) \phi(\ell - 2k\Delta - 2h)]\} \end{aligned}$$

where $\Delta = h - \ell$.

It follows from all of this that the joint density of $h_1, \ell_1, c_1, h_2, \ell_2$ is given by

$$p(h_1, \ell_1, c_1, h_2, \ell_2) = p(h_1, \ell_1, c_1) p(h_2 - c_1, \ell_2 - c_1) . \quad (4)$$

Since $\Delta_{12} = \max(h_1, h_2) - \min(\ell_1, \ell_2)$ is a function of h_1, h_2, ℓ_1, ℓ_2 we have

$$E[\Delta_{12}^2] = \int_{h_1, \ell_1, c_1, h_2, \ell_2} \Delta_{12}^2 p(h_1, \ell_1, c_1) p(h_2 - c_1, \ell_2 - c_1) dh_1 d\ell_1 dc_1 dh_2 d\ell_2 . \quad (5)$$

6 Simulations

We study the properties of the proposed High-Low Ratio (HLR) statistic and the Variance Ratio (VR) statistic in a simulation study. For each situation and sample size n (an even number of days), the number of simulated replications is 500, and the number of intraday periods is $NIP = 5000$. We consider several situations for the underlying data generating mechanism. In each situation, the underlying process is a geometric Brownian motion (gBm) with deviations as indicated. The base case is a gBm. Then we study a jump diffusion (JD) process, and a gBm to which we randomly add Bid-Ask noise (BA). All simulated log price processes are calculated over $NIP = 5000$ intraday periods for each of n days. For the calculation of VR , we compute n daily returns as the differences

of the daily log closing prices, and the two-day returns are the sums of the two one-day returns. The computation of HLR is based on non-overlapping two-day periods and one-day periods, that use all of the corresponding intraday log prices. The computation of VR is based on the same non-overlapping two-day periods and one-day periods, but uses only the closing log prices. We next describe the various situations in detail. For each situation, we begin the n -day period with a log price of zero, as the log closing price at the close of the zero'th trading day.

For simulating the gBM, the increments to the log price for each intraday period are *iid* normal with mean zero and standard deviation 0.001.

For simulating the JD, we start with log price increments as in the gBm case, and add jumps to a subset of these increments at times generated by a Poisson process. The jumps are iid normal with mean zero and standard deviation 0.05. The waiting time between jumps has a mean of 1000 intraday periods (that is, an expected value of 5 jumps per trading day).

For simulating the BA situation, we start by simulating the gBm as described above, and then we add to each intraday log price a noise process independent of the gBm and mutually independent taking values 0.005 and -0.005 with probabilities $1/2$ and $1/2$, respectively.

For each simulated realization, we compute Z_{VR} and Z_{HLR} and perform a z -test at the 5% significance level, two-tailed.

The following are some simulation results for $n = 500$. We report the proportion of the time that the null hypothesis is rejected by the z -test. For the base case of a gBm process, the proportion of rejections (representing the size of the test) is 6.2% for Z_{VR} and 5.6% for Z_{HLR} . We find that for this case, the means for VR and HLR are 0.998 and 1.007, and the standard deviations are 0.065 and 0.024, respectively. Notably, the standard deviation for VR is about 2.7 times larger than that for HLR, suggesting that the HLR is a more efficient way to test whether a process deviates from the gBm. We find similar efficiency gains for the other situations. For the non-gBM cases, the

proportion of rejections represents the power of the test. For the JD situation, the power is 5.6% for Z_{VR} and 54% for Z_{HLR} , respectively. In the BA situation, the power is 5.4% for Z_{VR} and 34% for Z_{HLR} , respectively. For a process that combines the JD and BA situations, the power is 7.4% for Z_{VR} and 18.2% for Z_{HLR} , respectively. It follows that in all cases studied, the power of the Z_{HLR} test greatly exceeds that of the Z_{VR} test.

7 Empirical estimations

We compare our HLR measure, based on high and low prices, and Lo and Mackinlay's (1998) return variance ratio, VR, for two prominent cryptocurrencies, Bitcoin and Ethereum, which are traded continuously, 24/7. The data source for our analysis is the daily price series from Finance.Yahoo.com. It provides for each day the high, low and closing prices denoted by PH_t , PL_t , PC_t , respectively. The analysis is conducted for both daily and weekly data, the week being from Wednesday to Tuesday.³ The daily log return is $r_{1,t} = \log PC_t - \log PC_{t-1}$, where PC_t is the closing (end-of-day) price on day t . The two-day log return is $r_{2,t} = r_{1,t} + r_{1,t-1}$. The sample includes n days, an even number. We then calculate s_{12} and s_{22} as, respectively, the sample variances of $r_{1,t}$, calculated from n days, and of $r_{2,t}$, calculated from $n/2$ non-overlapping even days, $r_{2,2}, r_{2,4}, r_{2,6}, \dots, r_{2,n}$. The "ratio of variances" (Lo and Mackinlay (1988, p. 46) is $VR = \frac{1}{2}s_{22}/s_{12}$. The test statistic based on VR is given by Lo and Mackinlay (1988, Equation (7)), $Z_{VR} = \sqrt{n/2}(VR - 1)$. For HLR, we use PH_t and PL_t , the high and low prices for day t . We define $H_t = PH_t$ and $L_t = PL_t$ and for each even period of two days that starts on day t , we define $HH_t = \max(H_t, H_{t-1})$ and $LL_t = \min(L_t, L_{t-1})$. HLR is given by

$$HLR = \frac{\text{Ave} \log^2(HH_t/LL_t)}{\text{Ave}[\log^2(H_t/L_t) + \log^2(H_{t-1}/L_{t-1})]} .$$

³In case of no trading on Tuesday, as it was after the 9/11/2001 attack.

$Ave[\cdot]$ is the average of the values in the brackets. The average in the numerator is calculated over $n/2$ non-overlapping two-period pairs for the even values $t = 2, 4, 6, \dots, n$. The test statistic based on HLR is given by $Z_{HLR} = \sqrt{n}(HLR - 1)/0.525$. For weekly data, these variables are calculated in a similar way. The log returns $r_{1,t}$ and $r_{2,t}$ are calculated for one week and for two non-overlapping weeks, respectively. The high and low prices H_t and L_t are the highest and lowest prices over all days of the week.

Data for Bitcoin is available for the period 10/3/2014 to 6/15/2024, 3,544 days and 1,772 two-day pairs, and data for Ethereum is available for the period 11/11/2017 to 6/14/2024, 2408 days and 1204 two-day pairs.

The results are presented in Table 1. Panels A1 and A2 present results based on daily data for Bitcoin and Ethereum, respectively, and in Panels B1 and B2 we present results for weekly data. In Panels A1 and A2 we present results for the entire period in Column (1), and for two equal subperiods in Columns (2) and (3). This enables us to observe whether the series characteristics have changed over time. The table also reports the slope coefficient $\hat{\rho}$ from a regression of r_t on r_{t-1} (and a constant)⁴ and its t -statistic, which employs a heteroskedastic-consistent standard error.

For both Bitcoin and Ethereum, HLR is significant and greater than 1.0 for the entire sample period (Column (1)). This indicates a significant deviation of the currencies' price processes from gBm. The difference $HLR - 1$ is greater in subperiod 1 than it is in subperiod 2, possibly indicating that as trading in cryptocurrencies became more active, their price process became closer to gBm. Over the two subperiods we find that HLR is significant and greater than 1.0 in the first subperiod (Column (2)), but in the second subperiod (Column (3)), the difference $HLR - 1$ is weakly significant (at $p = 0.078$) for Bitcoin and it is near zero for Ethereum.

For VR , we observe that for Bitcoin it is significant and greater than 1.0 for the entire sample

⁴This least-squares estimate is different from the first-order autocorrelation coefficient considered in Lo and MacKinlay (1988).

period (Column (1)) with the difference $VR - 1$ being weakly significant ($p = 0.061$). In subperiod 1 (Column (2)), VR is significant and greater than 1.0, while the least-squares estimated first-order autocorrelation coefficient $\hat{\rho}$ is near zero; Lo and MacKinlay (1988, p. 48) associate $VR - 1$ with the Pearson sample first-order autocorrelation and point out (p. 52) that their finding of $VR > 1$ for stock index returns is consistent with a positive return serial correlation for these return series. In Column (2), VR is significant and greater than 1.0 while $\hat{\rho}$ is practically zero; Lo and MacKinlay (1988) associate VR that is larger than 1.0 with a positive first-order return autocorrelation. For Ethereum, VR is significantly lower than 1.0 for the entire period and for each of the two subperiods while $\hat{\rho}$, which is negative, is insignificantly different from zero. When prices are sampled at a weekly frequency, we cannot reject the null hypothesis, either for VR or for HLR .

Table 1: Estimation results for cryptocurrencies: Bitcoin and Ethereum

Estimation results for two cryptocurrencies, Bitcoin and Ethereum. HLR is the average ratio of the squared logarithm of the high and low prices for each non-overlapping two-day period divided by the average of twice the ratio of the high and low prices for one-day periods. VR is the variance of two-day (log) returns divided by twice the variance of the one-day (log) returns. The analysis for weekly data is done in a similar way, with the unit time period being a week instead of a day. Z_{HLR} is the test statistic developed in this paper and Z_{VR} is the test statistic provided by Lo and MacKinlay (1988). The tests are for population HLR=1 and population VR=1. The statistic $\hat{\rho}$ is the slope coefficient from a regression of daily returns on lagged daily returns and a constant, and its t -statistic employs robust standard errors. ***, ** and * indicate significance at the 0.01, 0.05 and 0.10 levels, respectively (two-tailed tests).

Panel A: Daily prices and returns, using the following samples:

Bitcoin: 10/3/2014 to 6/15/2024, 3544 days. Ethereum: 11/11/2017 to 6/14/2024, 2408 days.

Variables	Entire period	Subperiod 1: 1st half	Subperiod 2: 2nd half
	(1)	(2)	(3)
Panel A1: Bitcoin			
HLR (Z_{HLR})	1.055 (6.26)***	1.083 (6.65)***	1.021 (1.76)*
VR (Z_{VR})	1.044 (1.87)*	1.085 (2.53)**	0.992 (-0.23)
$\hat{\rho}$ (t-statistic)	-0.022 (-0.82)	-0.002 (-0.04)	-0.049 (-1.49)
Panel A2: Ethereum			
HLR (Z_{HLR})	1.041 (3.81)***	1.069 (4.56)***	1.000 (-0.01)
VR (Z_{VR})	0.913 (-3.01)***	0.916 (-2.07)**	0.910 (-2.21)**
$\hat{\rho}$ (t-statistic)	-0.045 (-1.52)	-0.036 (-0.86)	-0.060 (-1.46)

Panel B: Weekly prices and returns, Wednesday to Tuesday.

	Panel B1: Bitcoin	Panel B2: Ethereum
HLR (Z -statistic)	1.013 (.056)	0.981 (-0.67)
VR (Z -statistic)	0.929 (-1.12)	0.924 (-1.03)
$\hat{\rho}$ (t -statistic)	0.019 (0.38)	0.008 (0.15)
n	504	342

In conclusion, we developed a new methodology for testing whether securities prices follow a geometric Brownian motion (gBm) and implemented it for testing whether the log prices of two major cryptocurrencies, Bitcoin and Ethereum, follow gBm. Importantly, these currencies are not subject to regulatory constraints that would inhibit their prices from reflecting what traders are willing to trade them at. We find that even in this unconstrained trading environment, the HLR tests suggest that the daily (24-hour) log prices deviate significantly from what would be expected under gBm. Interestingly, the log prices of both these currencies seem to have have conformed with gBm as their trading expanded: in the second subperiod of our sample we cannot reject the null hypothesis of prices following gBm. Prices also conform with the gBm when using weekly instead of daily time intervals.

8 References

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